Adaptive Dilation Factor Design Strategies for the Implicit Feature Extraction Module in SSSDTCN-based Industrial Time Series Imputation

Technical Research Plan

Abstract. The SSSDTCN model uses a dilated causal convolution stack as its implicit feature extraction module, with dilation rates fixed to the exponential schedule $\{1, 2, 4, 8, 16, 32\}$ before training. While this hand-crafted design achieves broad multi-scale temporal coverage, it cannot adapt to the specific periodicity or correlation structure of different industrial datasets or input segments. This document surveys and formalises **four principled approaches** for making the dilation factors *adaptive*: (I) learnable soft dilation parameters, (II) input-conditioned selective kernel fusion (SKNet-inspired), (III) frequency-spectral guided dilation, and (IV) temporal self-attention guided dynamic dilation. For each approach the source literature, core adaptive mechanism, and essential mathematical formulation are presented together with an illustrative diagram.

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Background

The implicit module of SSSDTCN stacks L dilated causal convolution layers. A 1-D dilated causal convolution with kernel $\mathbf{w}$, kernel size k, and dilation d computes:

$$(\mathbf{x} \star_d \mathbf{w})[t] = \sum_{i=0}^{k-1} \mathbf{w}[i] \cdot \mathbf{x}[t - d \cdot i], \tag{1}$$

enforcing causality via left-only zero-padding. The current model fixes $d_\ell = 2^{\ell-1}$, giving rates $\{1, 2, 4, 8, 16, 32\}$ for $L = 6$ and a total receptive field of $\mathrm{RF} = 1+(k-1)(2^L-1) = 253$ steps (for $k = 5$). Three limitations motivate adaptive designs: (1) the Debutanizer and SRU datasets have distinct dominant periodicities that a single fixed schedule cannot simultaneously optimise; (2) under high missing-ratio regimes longer context matters more, but the schedule is mask-blind; (3) non-stationary operating regimes cannot be tracked by static rates. Figure 1 illustrates the current fixed-dilation stack.

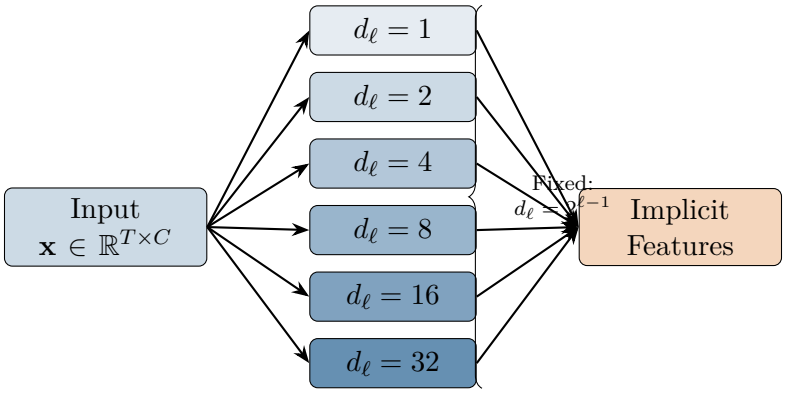

Figure 1: Current SSSDTCN implicit module: six dilated causal convolution layers with a fixed exponential dilation schedule.

1 Approach I: Learnable Soft Dilation Parameters

Reference: Dai et al. (2017), Deformable Convolutional Networks, ICCV; Chen et al. (2020), Temporal Deformable Convolution.

Inspired by deformable convolutions, which showed that spatial sampling offsets can be learned from data rather than fixed on a regular grid, this approach applies the same principle to the temporal dilation axis. Each layer ℓ is assigned an unconstrained trainable scalar $\lambda_\ell \in \mathbb{R}$; the effective dilation is $d_\ell = e^{\lambda_\ell}$, which is always positive and differentiable. For non-integer d_ℓ, linear interpolation between adjacent time steps enables gradient flow back to λ_ℓ. Because $\{\lambda_\ell\}$ are optimised jointly with all other model weights during end-to-end training, the dilation schedule converges to the values that best fit the training data rather than to a human-specified schedule. Once training ends the dilations are fixed; all inputs share the same learned rates.

Core equations.

$$d_\ell = \exp(\lambda_\ell), \tag{2}$$

$$(\mathbf{x} \star_{d_\ell} \mathbf{w})[t] = \sum_{i=0}^{k-1} \mathbf{w}[i] \cdot \underbrace{(1-\{d_\ell i\})\, \mathbf{x}[\lfloor t - d_\ell i \rfloor] + \{d_\ell i\}\, \mathbf{x}[\lceil t - d_\ell i \rceil]}_{\text{linear interpolation}}, \tag{3}$$

$$\mathcal{L} = \mathcal{L}_{\text{impute}} - \beta \sum_{\ell<\ell'} |d_\ell - d_{\ell'}|, \tag{4}$$

where $\{u\}$ denotes the fractional part of u and the diversity penalty eq. (4) (weight $\beta > 0$) prevents all layers collapsing to the same rate.

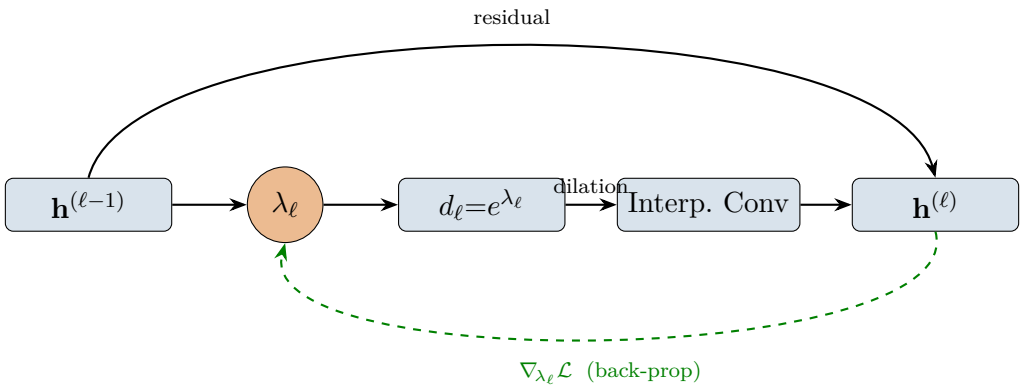

Figure 2: Approach I: $d_\ell = e^{\lambda_\ell}$ is a trainable scalar. The interpolated convolution passes gradients (green dashed arc) back to λ_ℓ, optimising the dilation end-to-end.

2 Approach II: Input-Conditioned Selective Kernel Dilation

Reference: Li et al. (2019), Selective Kernel Networks (SKNet), CVPR; Hu et al. (2018), Squeeze-and-Excitation Networks (SENet), CVPR.

Approach I's dilation rates are fixed after training and apply equally to every input. Approach II starts from a different premise: different inputs intrinsically require different dilation rates, so the rates should be determined by the current input rather than fixed in advance. The implementation proceeds as follows. K parallel dilated convolution branches with distinct candidate rates $\{d^{(1)}, \ldots, d^{(K)}\}$ are applied simultaneously to the same input, producing K feature maps. These are summed and globally average-pooled along the time axis, compressing the entire sequence into a fixed-length vector that captures the overall character of the current input. This vector is fed through two fully-connected layers to produce K selection weights (summing to 1, one per branch). Finally, the K feature maps are recombined as a weighted sum using those weights, yielding the layer output. Adaptivity arises because the weight vector is recomputed from scratch on every forward pass: a different input yields different weights, which in turn shifts the effective dilation toward the branch that best matches the dominant temporal scale of that input. This is the essential difference from Approach I—whereas Approach I's dilations are fixed at the end of training, Approach II's effective dilation changes with every new input at inference time.

Core equations.

$$\tilde{\mathbf{h}}^{(j)} = \mathrm{Conv}_{d^{(j)}}\Big(\mathbf{h}^{(\ell-1)}\Big), \quad j = 1, \ldots, K, \tag{5}$$

$$\mathbf{s} = \mathrm{AvgPool}\Big(\textstyle\sum_{j=1}^{K} \tilde{\mathbf{h}}^{(j)}\Big) \in \mathbb{R}^{H}, \tag{6}$$

$$\alpha^{(j)} = \mathrm{softmax}_j\Big(\mathbf{W}_2^{(j)}\, \mathrm{ReLU}(\mathbf{W}_1\, \mathbf{s})\Big), \tag{7}$$

$$\mathbf{h}^{(\ell)} = \sum_{j=1}^{K} \alpha^{(j)}\, \tilde{\mathbf{h}}^{(j)}. \tag{8}$$

The effective dilation is $d^{\mathrm{eff}} = \sum_j \alpha^{(j)} d^{(j)}$: different inputs yield different $\alpha^{(j)}$, shifting the effective rate.

Worked example. With K=3 branches ($d^{(1)}$=1, $d^{(2)}$=4, $d^{(3)}$=8), sequence length T=100, and hidden channels H=16: each branch independently processes the input and outputs a 100×16 feature map. Summing the three maps and averaging over the time axis yields a 16-dimensional descriptor $\mathbf{s}$; two FC layers followed by softmax convert $\mathbf{s}$ into three weights. For a rapidly oscillating input the weights might be $\boldsymbol{\alpha} = [0.70, 0.20, 0.10]$, keeping the effective dilation near 1; for a slowly varying input $\boldsymbol{\alpha} = [0.10, 0.30, 0.60]$ shifts dominance to d=8. The final output is the α-weighted sum of the three maps, still of shape 100×16.

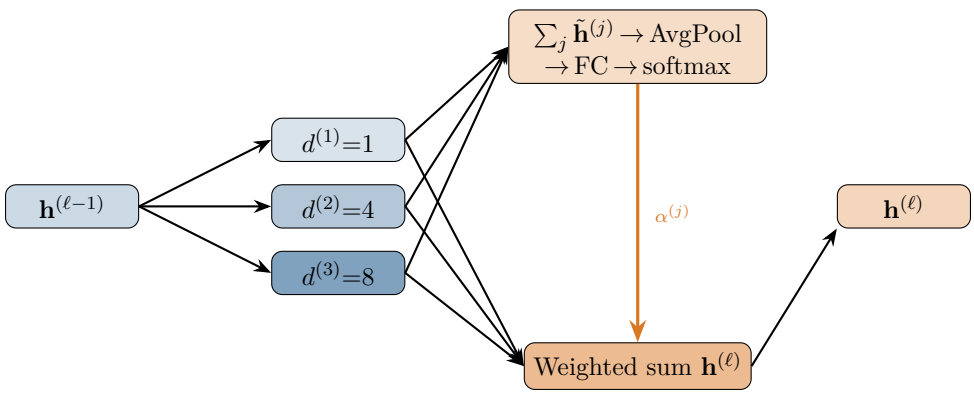

Figure 3: Approach II (K=3): three parallel branches (middle column) feed the *upper squeeze path* which computes input-conditioned weights $\alpha^{(j)}$ (orange arrow), and simultaneously feed the *lower combine path* for the final weighted sum.

3 Approach III: Frequency-Spectral Guided Adaptive Dilation

Reference: Wu et al. (2023), TimesNet, ICLR; Zhou et al. (2022), FEDformer, ICML.

The first two approaches rely on the training process to determine dilation rates. Approach III involves no training at all: it reads the appropriate dilation rates directly from the statistical properties of the data *before training begins*. Industrial time series typically contain several dominant periodic components. The core assumption is that a convolution kernel samples most efficiently when its stride equals one of those periods; under this assumption, the optimal dilation for layer ℓ is the one that makes the kernel's total sampling span equal to the ℓ-th dominant period. The procedure runs in three steps. First, the Fourier transform is applied to all channels of the training set and the

magnitude spectra are averaged across channels, producing a single spectrum that reflects the dataset's overall frequency structure. Second, the L largest peaks are identified, and each peak frequency is converted to a period length by dividing the sequence length by the frequency index. Third, for each period the required dilation is back-calculated from the target that the kernel's sampling span equals the period: $d_\ell = \lfloor P_\ell/(k-1) \rfloor$. These three steps execute once before training, and the resulting dilation schedule is installed as fixed layer configuration. No new parameters are introduced and no gradient updates are needed. The adaptivity is *dataset-level*: running the procedure on a different dataset yields a different schedule, but within one dataset the rates are fixed once computed. This is both the strength (zero training cost) and the limitation (cannot track non-stationarity within the dataset).

Core equations.

$$A_f = \frac{1}{C}\sum_{c=1}^{C}\Big|\mathrm{FFT}(\mathbf{x}[c,:])[f]\Big|, \quad f = 1, \ldots, \lfloor T/2 \rfloor, \tag{9}$$

$$f_1, \ldots, f_L = \mathrm{TopK}(\{A_f\},\, L)\,, \tag{10}$$

$$P_\ell = \lfloor T/f_\ell \rceil\,, \tag{11}$$

$$d_\ell = \max(1,\ \lfloor P_\ell/(k-1) \rfloor)\,. \tag{12}$$

Worked example. With T=480, k=5, L=3: suppose the three dominant FFT peaks fall at frequencies f=20, f=8, f=4. Applying eqs. (11) and (12):

Layer ℓ	Peak freq. f_ℓ	Period $P_\ell = T/f_\ell$ (steps)	Dilation $d_\ell = P_\ell/(k-1)$
1	20	24	6
2	8	60	15
3	4	120	30

Check: with d_1=6, k=5, the kernel samples positions t, $t-6$, $t-12$, $t-18$, $t-24$—a span of exactly 24 steps, matching the dominant period. Substituting a dataset whose top-3 peaks fall at different frequencies immediately yields a completely different dilation schedule, requiring no retraining.

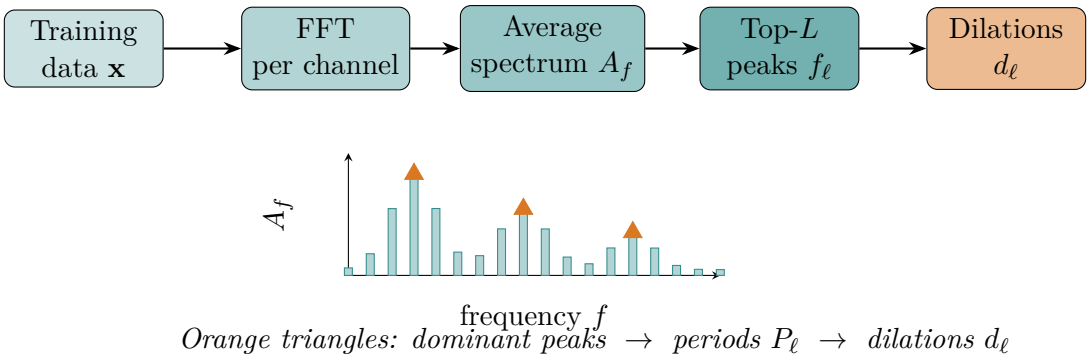

Figure 4: Approach III: the four-step offline calibration pipeline (top row). The average FFT magnitude spectrum (bottom) is computed once from the training set; the L tallest peaks (orange triangles) determine dominant periods, which are converted to dilation rates via eq. (12).

4 Approach IV: Temporal Self-Attention Guided Dynamic Dilation

Reference: Chen et al. (2020), Dynamic Convolution, CVPR; Wang et al. (2018), Non-local Neural Networks, CVPR.

The first three approaches share a common limitation: each convolution layer uses a single dilation rate applied uniformly to every position in the sequence. Approach IV breaks this restriction by computing a separate, position-specific effective dilation for every time step within a single forward pass. Before the convolution is applied, a lightweight self-attention module reads the layer's input and assigns to each position t a probability distribution over candidate offsets δ. The query vector for position t is dot-produced with the key vector of each historical position $t-\delta$; after softmax normalisation these dot-products become attention weights $a_{t,\delta}$. The expected offset $\bar{d}_t = \sum_\delta \delta\, a_{t,\delta}$ is the position's suggested dilation: when t is most similar to nearby history (high-frequency region), weights concentrate on small δ and $\bar{d}_t$ is small; when t relates more to distant history (low-frequency region), weights shift to large δ and $\bar{d}_t$ is large. A sigmoid gate then scales the base dilation $d_\ell^{(0)}$ by a context-dependent factor derived from $\mathbf{h}^{(\ell-1)}$, bounding the final rate within $[0,\, 2d_\ell^{(0)}]$ to prevent the receptive field from growing without bound. This computation is repeated independently for every position t, making Approach IV the finest-grained of all four methods. The cost is that the attention computation scales linearly with sequence length, making inference noticeably more expensive than the other approaches.

Core equations.

$$e_{t,\delta} = \frac{\mathbf{q}_t^\top \mathbf{k}_{t-\delta}}{\sqrt{d_k}}, \qquad a_{t,\delta} = \frac{\exp(e_{t,\delta})}{\sum_{\delta'} \exp(e_{t,\delta'})}, \tag{13}$$

$$\bar{d}_t = \sum_{\delta=0}^{T_{\max}} \delta \cdot a_{t,\delta}, \tag{14}$$

$$\tilde{d}_{t,\ell} = d_\ell^{(0)} \cdot 2\ \mathrm{sigmoid}\Big(\mathbf{w}_s^\top \mathbf{h}^{(\ell-1)}[:,t] + b_s\Big), \tag{15}$$

where eq. (15) constrains the adapted dilation to $[0,\, 2d_\ell^{(0)}]$.

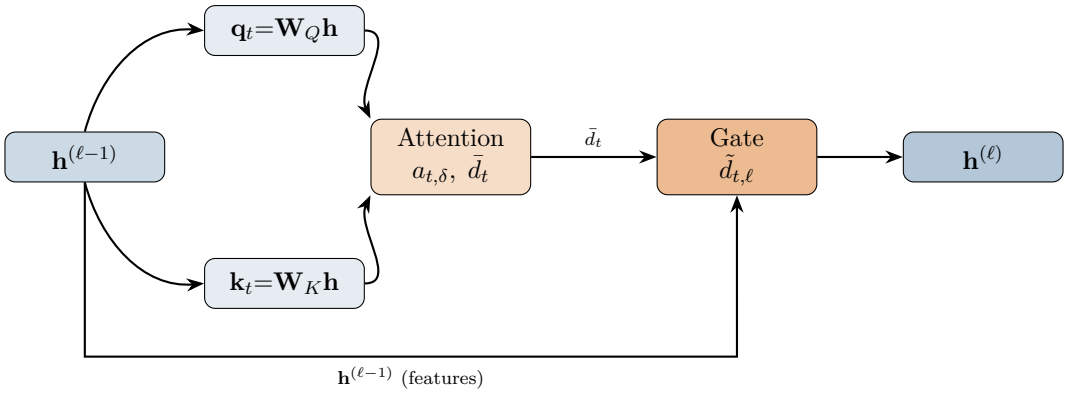

Figure 5: Approach IV: $\mathbf{h}^{(\ell-1)}$ is projected to query $\mathbf{q}_t$ (upper branch) and key $\mathbf{k}_t$ (lower branch); their dot products yield per-offset attention weights $a_{t,\delta}$, whose expectation gives the suggested dilation $\bar{d}_t$. The sigmoid gate (right) additionally takes the raw hidden state (bottom path) and scales $d_\ell^{(0)}$ to produce $\tilde{d}_{t,\ell} \in [0,\, 2d_\ell^{(0)}]$.

5 Comparative Discussion

Table 1 summarises the four approaches along key design dimensions. For a practical starting point, a hybrid of Approaches III and I is recommended: use the spectral calibration (eq. (12)) to initialise $\lambda_\ell^{(0)} = \log(d_\ell^{\text{spectral}})$, then fine-tune with learnable soft dilations. This combines physics-guided initialisation with end-to-end refinement at minimal additional cost.

Table 1: Comparison of adaptive dilation approaches.

	Approach	**Extra params**	**Adapt. level**	**Training needed?**	**Inference cost**
—	Fixed (baseline)	0	—	No (preset)	$\mathcal{O}(LC)$
I	Soft dilation	L scalars	Global	Yes (E2E)	$\mathcal{O}(LC)$
II	SKNet fusion	$O(KH^2)$	Per-sample	Yes (E2E)	$\mathcal{O}(KLC)$
III	Spectral calibration	0	Per-dataset	No (offline)	$\mathcal{O}(LC)$
IV	Attn-guided	$O(Hd_kL)$	Per-position	Yes (E2E)	$\mathcal{O}(LCT)$